

Failed Authentication Detection and Investigation using Windows Security Logs and Wazuh

Objective

The objective of this simulation was to investigate how Windows Security Logs and Wazuh detect failed authentication attempts involving invalid usernames, incorrect passwords, and repeated logon failures commonly associated with account enumeration, credential guessing, and brute-force activity.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	Brute Force	T1110
Discovery	Account Discovery	T1087
Initial Access	Valid Accounts	T1078

Attack Simulation

The following authentication failure scenarios were generated:

Scenario 1 — Invalid Username Authentication Attempts

Attempted logon using:

- fake username
- incorrect password

Repeated multiple times.

Scenario 2 — Valid Username with Incorrect Password

Attempted logon using:

- valid local account (socadmin)
- incorrect password

Repeated multiple times.

Attack Chain Overview

Step	Activity
1	User authentication attempt initiated
2	Windows credential validation performed
3	Authentication failed
4	Security Event ID 4625 generated
5	Wazuh collected failed authentication telemetry
6	Repeated failures increased behavioral suspicion

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> runas /user:fakeuser cmd
Enter the password for fakeuser:
Attempting to start cmd as user "DESKTOP-HDF09SQ\fakeuser" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

PS C:\WINDOWS\system32> runas /user:fakeuser cmd
Enter the password for fakeuser:
Attempting to start cmd as user "DESKTOP-HDF09SQ\fakeuser" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

PS C:\WINDOWS\system32> runas /user:fakeuser cmd
Enter the password for fakeuser:
Attempting to start cmd as user "DESKTOP-HDF09SQ\fakeuser" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

PS C:\WINDOWS\system32> runas /user:fakeuser cmd
Enter the password for fakeuser:
Attempting to start cmd as user "DESKTOP-HDF09SQ\fakeuser" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

PS C:\WINDOWS\system32> runas /user:socadmin cmd
Enter the password for socadmin:
Attempting to start cmd as user "DESKTOP-HDF09SQ\socadmin" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

PS C:\WINDOWS\system32> _
```

Figure 1 — Failed authentication attempts generated in the lab environment

Windows Security Log Investigation

Key Security Events Identified

Event ID	Detection Purpose
4625	Failed Logon Attempt
4624	Success Logon Attempt

Important Authentication Failure Types Observed

Scenario	Behavioral Meaning
Invalid username	Enumeration / reconnaissance behavior
Valid username + bad password	Credential guessing / brute force behavior

Windows Event ID 4625 — Failed Authentication Analysis

Key Findings

Field	Value
Event ID	4625
Logon Type	2 (Interactive)
Authentication Package	Negotiate
Process Name	svchost.exe
Source Address	::1
Workstation	DESKTOP-HDF09SQ

Investigation Analysis

Windows Security Event ID 4625 captured failed authentication attempts generated from the local system.

The authentication attempts included:

- invalid account names
- incorrect passwords
- repeated failed login behavior

The repeated authentication failures increased behavioral suspicion because repeated login failures are commonly associated with:

- account enumeration
- password guessing
- brute-force activity
- credential validation attempts

The investigation also confirmed that all attempts originated from:

- localhost (::1)
- interactive logon activity
- local workstation activity

which reduced overall incident severity.

Important SOC Analyst Insight

A single failed login attempt is often:

- user error
- mistyped credentials
- normal operational activity

However, repeated failed authentication attempts significantly increase concern level, especially when involving:

- multiple usernames
- administrator accounts
- remote systems
- domain accounts
- rapid authentication sequences

Behavioral context is more important than isolated events.

Invalid Username Investigation

Scenario 1 — Fake Username Authentication Failure

Important Findings

Field	Value
TargetUserName	fakeuser
Status	0xc000006d
SubStatus	0xc0000064
Logon Type	2
Source Address	::1

Failure Reason Analysis

Status Code

0xc000006d

Meaning:

General authentication failure

SubStatus Code

0xc0000064

Meaning:

Username does not exist

Investigation Analysis

The failed authentication attempts targeting the username:

fakeuser

indicated invalid account authentication attempts against a non-existent user.

This behavior is commonly associated with:

- username enumeration
- account discovery
- reconnaissance activity
- brute-force preparation

The repeated failures increased behavioral suspicion because attackers frequently probe systems for valid usernames prior to password attacks.

However, investigation confirmed:

- localhost-only activity
- interactive logon attempts
- lab-generated testing behavior

which reduced incident severity.

Security Number of events: 9,494				
Keywords	Date and Time	Source	Event ID	Task Category
Audit Failure	5/24/2026 3:55:08 PM	Microsoft Windows security auditing.	4625	Logon
Audit Success	5/24/2026 3:55:07 PM	Microsoft Windows security auditing.	5379	User Account Management
Audit Failure	5/24/2026 3:53:28 PM	Microsoft Windows security auditing.	4625	Logon
Audit Success	5/24/2026 3:52:34 PM	Microsoft Windows security auditing.	5379	User Account Management
Audit Success	5/24/2026 3:52:29 PM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	5/24/2026 3:52:29 PM	Microsoft Windows security auditing.	4624	Logon
Audit Success	5/24/2026 3:50:40 PM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	5/24/2026 3:50:40 PM	Microsoft Windows security auditing.	4624	Logon

Event 4625, Microsoft Windows security auditing.	
General	Details
☒ Friendly View ☐ XML View	
SubjectUserName socadmin	
SubjectDomainName DESKTOP-HDF09SQ	
SubjectLogonId 0x5023a	
TargetUserSid S-1-0-0	
TargetUserName fakeuser	
TargetDomainName DESKTOP-HDF09SQ	
Status 0xc000006d	
FailureReason %2313	
SubStatus 0xc0000064	
LogonType 2	
LogonProcessName seclogon	
AuthenticationPackageName Negotiate	
WorkstationName DESKTOP-HDF09SQ	
TransmittedServices -	
LmPackageName -	
KeyLength 0	
ProcessId 0x279c	
ProcessName C:\Windows\System32\svchost.exe	
IpAddress ::1	

Figure 2 — Failed authentication event showing invalid username enumeration behavior

Valid Username + Incorrect Password Investigation

Scenario 2 — Credential Guessing Activity

Important Findings

Field	Value
TargetUserName	socadmin
Status	0xc000006d
SubStatus	0xc000006a
Authentication Package	Negotiate
Logon Process	seclogo

Failure Reason Analysis

SubStatus Code

0xc000006a

Meaning:

Correct username, wrong password

Important SOC Insight

This scenario was operationally different from invalid username enumeration because the authentication attempts targeted a valid local account.

This significantly increased suspicion because it resembled:

- password guessing
- brute-force activity

- credential attack behavior
- attempted account compromise

Attackers commonly transition from:

- username discovery
to
- credential attacks

during early intrusion stages.

Investigation Analysis

The account:

```
socadmin
```

was confirmed to exist locally on the system.

Repeated failed authentication attempts against a valid account significantly increased behavioral concern because attackers commonly target:

- administrator accounts
- privileged users
- known valid usernames

during password spraying and brute-force operations.

The observed:

- localhost source address
- interactive logon type
- lab-generated activity

prevented escalation during investigation.

Security Number of events: 9,498				
Keywords	Date and Time	Source	Event ID	Task Category
Audit Failure	5/24/2026 4:00:44 PM	Microsoft Windows security auditing.	4625	Logon
Audit Success	5/24/2026 4:00:42 PM	Microsoft Windows security auditing.	5379	User Account Management
Audit Success	5/24/2026 3:59:34 PM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	5/24/2026 3:59:34 PM	Microsoft Windows security auditing.	4624	Logon
Audit Success	5/24/2026 3:55:56 PM	Microsoft Windows security auditing.	5379	User Account Management
Audit Success	5/24/2026 3:55:56 PM	Microsoft Windows security auditing.	5379	User Account Management
Audit Success	5/24/2026 3:55:56 PM	Microsoft Windows security auditing.	5379	User Account Management
Audit Success	5/24/2026 3:55:56 PM	Microsoft Windows security auditing.	5379	User Account Management

Event 4625, Microsoft Windows security auditing.	
General	Details
☒ Friendly View ☐ XML View	
SubjectUserName socadmin SubjectDomainName DESKTOP-HDF09SQ SubjectLogonId 0x5023a TargetUserSid S-1-0-0 TargetUserName socadmin TargetDomainName DESKTOP-HDF09SQ Status 0xc000006d FailureReason %2313 SubStatus 0xc000006a LogonType 2 LogonProcessName seclogon AuthenticationPackageName Negotiate WorkstationName DESKTOP-HDF09SQ TransmittedServices - LmPackageName - KeyLength 0 ProcessId 0x279c ProcessName C:\Windows\System32\svchost.exe IpAddress ::1	

Figure 3 — Failed authentication event showing credential guessing behavior against a valid account

Logon Type Analysis

Interactive Logon Investigation

Logon Type Observed

Logon Type: 2

Meaning:

Interactive local logon

Investigation Analysis

The authentication attempts used:

- local keyboard interaction
- interactive logon behavior
- local authentication processing

This indicated:

- local activity
- not RDP-based attacks
- not SMB/network logons
- not remote authentication traffic

Understanding logon types is critical during authentication investigations because:

- remote logons significantly increase severity
- network logons may indicate lateral movement
- RDP logons may indicate external compromise attempts

Important Authentication Failure Codes

SubStatus Code	Meaning
0xc0000064	Username does not exist
0xc000006a	Incorrect password
0xc0000234	Account locked
0xc0000072	Account disabled

Wazuh Detection Analysis

Wazuh Detection Sources

Detection Source	Purpose
Windows Security Logs	Failed authentication monitoring
Event ID 4625 Collection	Authentication failure telemetry
Behavioral Correlation	Repeated failed logon activity

Investigation Analysis

Wazuh successfully collected and centralized Windows authentication failure telemetry generated from repeated failed login attempts.

The investigation demonstrated how:

- repeated failed logons
- invalid usernames
- credential guessing attempts
- valid account targeting

can provide valuable indicators during:

- account enumeration investigations
- brute-force investigations
- credential attack analysis

The strongest behavioral indicators included:

- repeated authentication failures
 - targeting of valid accounts
 - sequential failed logons
 - authentication failure reason codes
-

Important False Positive Analysis

Failed authentication events are extremely common in enterprise environments and frequently occur due to:

- mistyped passwords
- expired credentials
- cached authentication failures
- user operational mistakes

The strongest indicators during this investigation were:

- repeated execution
- authentication sequencing

- username targeting behavior
- behavioral context

rather than the failed login event alone.

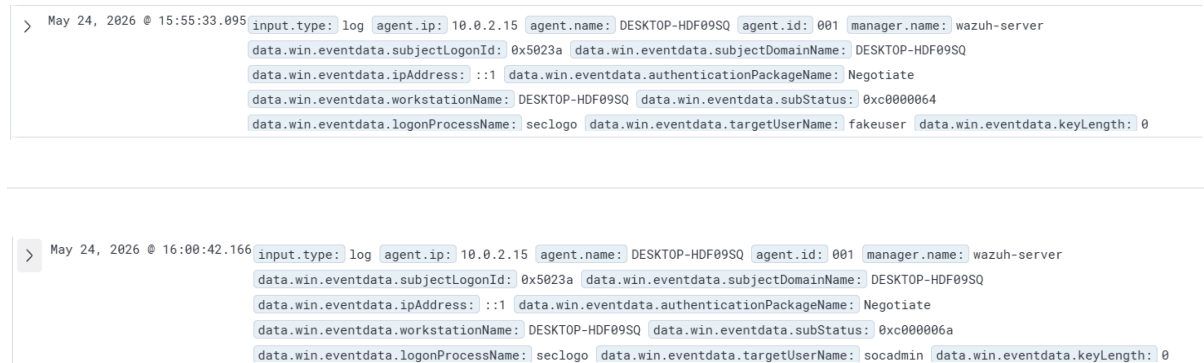


Figure 6 — Wazuh dashboard showing failed authentication telemetry

```

{
  "_index": "wazuh-alerts-4.x-2026.05.24",
  "_id": "PM6EWZ4Bsdig1Cmu3MWa",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "10.0.2.15",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "subjectLogonId": "0x5023a",
          "subjectDomainName": "DESKTOP-HDF09SQ",
          "ipAddress": "::1",
          "authenticationPackageName": "Negotiate",

```

```
"workstationName": "DESKTOP-HDF09SQ",
"subStatus": "0xc0000064",
"logonProcessName": "seclogo",
"targetUserName": "fakeuser",
"keyLength": "0",
"subjectUserSid": "S-1-5-21-221611468-1794481200-2977160085-1001",
"processId": "0x279c",
"processName": "C:\\\\Windows\\\\System32\\\\svchost.exe",
"ipPort": "0",
"failureReason": "%%2313",
"targetDomainName": "DESKTOP-HDF09SQ",
"targetUserSid": "S-1-0-0",
"logonType": "2",
"subjectUserName": "socadmin",
"status": "0xc000006d"
},
"system": {
"eventId": "4625",
"keywords": "0x8010000000000000",
"providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}",
"level": "0",
"channel": "Security",
"opcode": "0",
"message": "\"An account failed to log on.\\r\\n\\r\\nSubject:\\r\\n\\tSecurity ID:\\t\\tS-1-5-21-221611468-1794481200-2977160085-1001\\r\\n\\tAccount Name:\\t\\tsocadmin\\r\\n\\tAccount Domain:\\t\\tDESKTOP-HDF09SQ\\r\\n\\tLogon ID:\\t\\t0x5023A\\r\\n\\r\\nLogon Type:\\t\\t\\t2\\r\\n\\r\\nAccount For Which Logon Failed:\\r\\n\\tSecurity ID:\\t\\tS-1-0-0\\r\\n\\tAccount Name:\\t\\tfakeuser\\r\\n\\tAccount Domain:\\t\\tDESKTOP-HDF09SQ\\r\\n\\r\\nFailure Information:\\r\\n\\tFailure Reason:\\t\\tUnknown user name or bad password.\\r\\n\\tStatus:\\t\\t\\t0xC000006D\\r\\n\\tSub Status:\\t\\t0xC0000064\\r\\n\\r\\nProcess Information:\\r\\n\\tCaller Process ID:\\t0x279c\\r\\n\\tCaller Process Name:\\tC:\\\\Windows\\\\System32\\\\svchost.exe\\r\\n\\r\\nNetwork Information:\\r\\n\\tWorkstation Name:\\tDESKTOP-HDF09SQ\\r\\n\\tSource Network Address:\\t::1\\r\\n\\tSource Port:\\t\\t0\\r\\n\\r\\nDetailed Authentication Information:\\r\\n\\tLogon Process:\\t\\tseclogo\\r\\n\\tAuthentication Package:\\tNegotiate\\r\\n\\tTransited Services:\\t-\\r\\n\\tPackage Name (NTLM
```

only):\t-\r\n\tKey Length:\t\t0\r\n\r\nThis event is generated when a logon request fails. It is generated on the computer where access was attempted.\r\n\r\nThe Subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.\r\n\r\nThe Logon Type field indicates the kind of logon that was requested. The most common types are 2 (interactive) and 3 (network).\r\n\r\nThe Process Information fields indicate which account and process on the system requested the logon.\r\n\r\nThe Network Information fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.\r\n\r\nThe authentication information fields provide detailed information about this specific logon request.\r\n\r\n\t- Transited services indicate which intermediate services have participated in this logon request.\r\n\r\n\t- Package name indicates which sub-protocol was used among the NTLM protocols.\r\n\r\n\t- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.\\"",

```
"version": "0",
"systemTime": "2026-05-24T10:25:08.9447100Z",
"eventRecordID": "9482",
"threadID": "4628",
"computer": "DESKTOP-HDF09SQ",
"task": "12544",
"processID": "900",
"severityValue": "AUDIT_FAILURE",
"providerName": "Microsoft-Windows-Security-Auditing"
}
},
"rule": {
"mail": false,
"level": 5,
"pci_dss": [
"10.2.4",
"10.2.5"
],
"hipaa": [
"164.312.b"
```

```
],
"tsc": [
  "CC6.1",
  "CC6.8",
  "CC7.2",
  "CC7.3"
],
"description": "Logon Failure - Unknown user or bad password",
"groups": [
  "windows",
  "windows_security",
  "authentication_failed"
],
"nist_800_53": [
  "AU.14",
  "AC.7"
],
"gdpr": [
  "IV_35.7.d",
  "IV_32.2"
],
"firedtimes": 7,
"mitre": {
  "technique": [
    "Account Access Removal"
  ],
  "id": [
    "T1531"
  ],
  "tactic": [
    "Impact"
  ]
},
"id": "60122",
"gpg13": [
  "7.1"
],
},
```



```
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779618333.2829984",
"timestamp": "2026-05-24T10:25:33.095+0000"
},
"fields": {
"timestamp": [
"2026-05-24T10:25:33.095Z"
]
},
"sort": [
1779618333095
]
}
```

```
{
"_index": "wazuh-alerts-4.x-2026.05.24",
"_id": "Vc6JWZ4Bsdig1CmumMUI",
"_version": 1,
"_score": null,
"_source": {
"input": {
"type": "log"
},
"agent": {
"ip": "10.0.2.15",
"name": "DESKTOP-HDF09SQ",
"id": "001"
},
"manager": {
"name": "wazuh-server"
},
"data": {
"win": {
"eventdata": {
"subjectLogonId": "0x5023a",
"subjectDomainName": "DESKTOP-HDF09SQ",
```

```
"ipAddress": "::1",
"authenticationPackageName": "Negotiate",
"workstationName": "DESKTOP-HDF09SQ",
"subStatus": "0xc000006a",
"logonProcessName": "seclogo",
"targetUserName": "socadmin",
"keyLength": "0",
"subjectUserSid": "S-1-5-21-221611468-1794481200-2977160085-1001",
"processId": "0x279c",
"processName": "C:\\\\Windows\\\\System32\\\\svchost.exe",
"ipPort": "0",
"failureReason": "%%2313",
"targetDomainName": "DESKTOP-HDF09SQ",
"targetUserSid": "S-1-0-0",
"logonType": "2",
"subjectUserName": "socadmin",
"status": "0xc000006d"
},
"system": {
"eventId": "4625",
"keywords": "0x8010000000000000",
"providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}",
"level": "0",
"channel": "Security",
"opcode": "0",
"message": "\"An account failed to log on.\\r\\n\\r\\nSubject:\\r\\n\\tSecurity ID:\\t\\tS-1-5-21-221611468-1794481200-2977160085-1001\\r\\n\\tAccount Name:\\t\\tsocadmin\\r\\n\\tAccount Domain:\\t\\tDESKTOP-HDF09SQ\\r\\n\\tLogon ID:\\t\\t0x5023A\\r\\n\\r\\nLogon Type:\\t\\t2\\r\\n\\r\\nAccount For Which Logon Failed:\\r\\n\\tSecurity ID:\\t\\tS-1-0-0\\r\\n\\tAccount Name:\\t\\tsocadmin\\r\\n\\tAccount Domain:\\t\\tDESKTOP-HDF09SQ\\r\\n\\r\\nFailure Information:\\r\\n\\tFailure Reason:\\t\\tUnknown user name or bad password.\\r\\n\\tStatus:\\t\\t\\t0xC000006D\\r\\n\\tSub Status:\\t\\t0xC000006A\\r\\n\\r\\nProcess Information:\\r\\n\\tCaller Process ID:\\t0x279c\\r\\n\\tCaller Process Name:\\tC:\\\\Windows\\\\System32\\\\svchost.exe\\r\\n\\r\\nNetwork Information:\\r\\n\\tWorkstation Name:\\tDESKTOP-HDF09SQ\\r\\n\\tSource Network Address:\\t::1\\r\\n\\tSource Port:\\t0\\r\\n\\r\\nDetailed Authentication
```

Information:\r\n\tLogon Process:\t\tseclogo\r\n\tAuthentication Package:\tNegotiate\r\n\tTransited Services:\t-\r\n\tPackage Name (NTLM only):\t-\r\n\tKey Length:\t\t0\r\n\r\nThis event is generated when a logon request fails. It is generated on the computer where access was attempted.\r\n\r\nThe Subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.\r\n\r\nThe Logon Type field indicates the kind of logon that was requested. The most common types are 2 (interactive) and 3 (network).\r\n\r\nThe Process Information fields indicate which account and process on the system requested the logon.\r\n\r\nThe Network Information fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.\r\n\r\nThe authentication information fields provide detailed information about this specific logon request.\r\n\r\n\t- Transited services indicate which intermediate services have participated in this logon request.\r\n\r\n\t- Package name indicates which sub-protocol was used among the NTLM protocols.\r\n\r\n\t- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.\r\n",

```

"version": "0",
"systemTime": "2026-05-24T10:30:44.6788949Z",
"eventRecordID": "9498",
"threadID": "308",
"computer": "DESKTOP-HDF09SQ",
"task": "12544",
"processID": "900",
"severityValue": "AUDIT_FAILURE",
"providerName": "Microsoft-Windows-Security-Auditing"
}
},
"rule": {
"mail": false,
"level": 5,
"pci_dss": [
"10.2.4",
"10.2.5"
],

```

```
"hipaa": [
  "164.312.b"
],
"tsc": [
  "CC6.1",
  "CC6.8",
  "CC7.2",
  "CC7.3"
],
"description": "Logon Failure - Unknown user or bad password",
"groups": [
  "windows",
  "windows_security",
  "authentication_failed"
],
"nist_800_53": [
  "AU.14",
  "AC.7"
],
"gdpr": [
  "IV_35.7.d",
  "IV_32.2"
],
"firedtimes": 8,
"mitre": {
  "technique": [
    "Account Access Removal"
  ],
  "id": [
    "T1531"
  ],
  "tactic": [
    "Impact"
  ]
},
"id": "60122",
"gpg13": [
  "7.1"
```

```

]
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779618642.2862158",
"timestamp": "2026-05-24T10:30:42.166+0000"
},
"fields": {
"timestamp": [
"2026-05-24T10:30:42.166Z"
]
},
"sort": [
1779618642166
]
}

```

Figure 7 — Detailed Wazuh telemetry for failed authentication activity

Attack Timeline

Time	Event
3:55 AM	Invalid username authentication attempt initiated
3:55:08 PM	Event ID 4625 generated for fakeuser
3:55:20 PM	Multiple failed attempts repeated
4:00:30 PM	Valid account authentication attempt initiated
4:00:44 PM	Event ID 4625 generated for socadmin
4:00:42 PM	Wazuh collected authentication telemetry

Key Detection Findings

Finding	Impact
Invalid username failures	Possible enumeration activity

Finding	Impact
Valid username + wrong password	Credential guessing behavior
Repeated failed logons	Increased behavioral suspicion
Interactive logon activity	Local authentication attempts
Localhost source address	Reduced incident severity
Authentication failure codes	Improved investigation accuracy
Behavioral correlation	Stronger SOC analysis

Primary Investigation Artifacts

Windows Security Event IDs

Event ID	Meaning
4625	Failed Logon Attempt
4624	Success Logon Attempt

Windows Logon Types

Logon Type	Meaning	Why Important
2	Interactive logon	Local user activity
3	Network logon	SMB/lateral movement
5	Service logon	Service persistence
9	NewCredentials (RunAs)	RunAs credential usage
10	RemoteInteractive (RDP)	RDP attacks and remote compromise

Important Authentication Fields

Field	Investigation Value
TargetUserName	Account being targeted
LogonType	Authentication method
Status	General failure reason
SubStatus	Exact authentication failure

Field	Investigation Value
IpAddress	Source location
AuthenticationPackageName	NTLM/Kerberos usage
LogonProcessName	Authentication mechanism

Conclusion

This simulation demonstrated how Windows Security Logs and Wazuh provide strong visibility into failed authentication activity involving account enumeration, invalid username attempts, and credential guessing behavior.

Behavioral correlation between:

- repeated failed logons
- username targeting
- authentication failure codes
- logon types
- source analysis
- account validation behavior

significantly improved investigation quality and detection confidence.

The investigation also highlighted the importance of contextual authentication analysis because the strongest indicators were not the failed logon events themselves, but the behavioral patterns surrounding:

- repeated failures
- valid account targeting
- authentication sequencing
- credential attack behavior
- reconnaissance activity

which together provide significantly stronger indicators of suspicious authentication activity.